Languages as Mechanisms of Interaction: The case of weak object pronouns

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- The problem of clitics
- An intro to the DS account of clitics and clitic related phenomena

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  - Turn taking at the sublexical level?
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Some prototypical examples

(1) Maria lo ama
Maria him.CL-ACC loves
Maria loves him [Italian]

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Maria him.CL-ACC sees
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(7) Maria lo ama
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(9) O Giorgos ton kseri
the.NOM George.NOM him.CL-ACC knows
George knows him [Standard Modern Greek (SMG)]

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Properties of clitics - Positioning systems

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    Proclisis with finite verbs
    Enclisis with non-finites

    (16)

To	katalava	(*to)	diavazontas to	
it.CL-ACC	understood	it.CL-ACC	reading	it.CL-ACC

I understood it by reading it [Standard Modern Greek]

Properties of clitics - Positioning systems

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  Enclisis as default
  - Proclisis with function word/stressed words/subordinators at left periphery

(20)

Poli anthropi (*to) kamnun to sosta
many people it.CL-ACC do.3PL it.CL-ACC right
Many people people do this right [Cypriot Greek]

(21)

Pios/En ton ide (*ton)?
who/NEG him.CL-ACC saw.3SG him.CL-ACC
Who saw him?/S/He did not see him [Cypriot Greek]

(22)

Otan/Ean ton ide (*ton)...
when/if him.CL-ACC saw.3SG him.CL-ACC
When/if s/he saw him [Cypriot Greek]

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(35)

(*Aton) Ekser aton
him.ACC know.1SG him.ACC
'I know him.'

(36)

(*Me) Δos me avuto to vivlio
me.ACC give.2SG.IMP me.ACC this the book
'Give me this book.'

(37)

Pios (*aton) entoke(n) aton?
who him.CL hit.1SG him.CL
'Who hit him?'

(38)

Ki	(*aton)	kser	aton
NEG	him.ACC	know.1SG	him.ACC

'I do not know him.

(39)

O	GIORIKAS	(*aton)	kser	aton
the.NOM	George.NOM	him.ACC	know.1SG	him.ACC

'It is George that knows him.'

Properties of clitics - Doubling

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(50)

Ton	Giorgo(,)	ton	ksero
the.ACC	George.ACC	him.CL-ACC	know.1SG

'I know George.' [Standard Modern Greek]

(51)

Ton	ksero	ton	Giorgo
him.CL-ACC	know.1SG	the.ACC	George.ACC

'I know George.' [Standard Modern Greek]

Properties of clitics - Clitic Climbing

The clitic attaches to a higher verb that does not consitute an argument of (Rizzi 1977, Wurmbrand 1998, Cinque 2006, Cardinaletti & Shlonsky, 2004)

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- Climbing Languages: only with modal, aspectual and motion verbs

Properties of clitics - Clitic Climbing

(52)

Maria lo puo comprare(lo)
Maria it.CL-ACC wants buy.INF
'Maria can to buy it.' [Italian]

(53)

Lo finisco di vedere(lo)
it.CL-ACC finish COMP buy.INF
'I finish seeing it.' [Italian]

(54)

Lo vado a comprare(lo)
it.CL-ACC go to buy.INF
'I'm going in order to buy it.' [Italian]

The Person Case Constraint

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(61)
*Le me ha dado
him.CL-DAT me.CL has given
'He/She has given me to him.' [Spanish]

(62)
*Paul me lui présentera
Paul me.CL him.CL-DAT will-introduce
'Paul will introduce me to him.' [French]

(63)
*Mi ti presento
me.CL him.CL introduce
'Paul introduces me to him.' [Italian]

DS account of clitics-Positioning

- Positioning systems: encoding positioning as restrictions on parsing triggers

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- The entry for 3rd person accusative clitic *to* "it" in SMG

IF	$?Ty(t), Tn(a)$	
THEN	IF	$[\downarrow_1^+]?Ty(x)$
	OR	
	IF	$Mood(-FINITE)$
	THEN	$\texttt{make}(\langle\downarrow_1\rangle); \texttt{go}(\langle\downarrow_1\rangle);$
		$\texttt{make}(\langle\downarrow_0\rangle); \texttt{go}(\langle\downarrow_0\rangle)$
		$\texttt{put}(Ty(e), Fo(U_{neut'}), ?\exists \mathbf{x}.Fo(\mathbf{x}));$
		gofirst(?Ty(t))
ELSE	Abort	

DS account of clitics-Positioning

- Cypriot Greek positioning in DS (Chatzikyriakidis 2010)
  - Earlier encoding of pragmatic preferences as syntactic triggers (we argue via routinization (Pickering & Garrod 1994, Bouzouita 2008 for Spanish))

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IF		$?Ty(t)$
THEN	IF	$\langle\downarrow_*\rangle ?\exists \mathbf{x}.Tn(\mathbf{x})\vert$
		$\langle\downarrow_0\rangle ?Ty(e_s)\vert$
		$\langle\downarrow_1^+\rangle Ty(x)$
	THEN	ACTIONS
ELSE		abort
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  - Clitics appeared as early as possible once a propositional domain has been established (Cann & Kempson 2008)
  - The actual strategy for parsing the individual elements got routinized and stored as a triggering restriction (Bouzouita 2008, Chatzikyriakidis, Gibson 2012 for Rangi)

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Person Case Constraint in DS

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Dative and 1st/2nd person clitics in Spanish/Italian syncretic project locally unfixed node

IF	$?Ty(t), Tn(a)$	
THEN	IF	$[\downarrow_1^+]?Ty(x)$
	OR	
	IF	$Mood(Imp)$
	THEN	$\mathtt{make}(\langle\downarrow_1^*\rangle); \mathtt{go}(\langle\downarrow_1^*\rangle);$
		$\mathtt{make}(\langle\downarrow_0\rangle); \mathtt{go}(\langle\downarrow_0\rangle)$
		$\mathtt{put}(\langle\uparrow_0\rangle\langle\uparrow_1^*\rangle Tn(a));$
		$\mathtt{make}(\langle\downarrow_0\rangle); \mathtt{go}(\langle\downarrow_0\rangle)$
		$\mathtt{put}(Ty(e), Fo(U_{x'}), ?\exists\mathbf{x}.Fo(\mathbf{x}))$
		$\mathtt{put}(?\exists\mathbf{x}.Tn(\mathbf{x}), gofirst(?Ty(t)))$
ELSE	Abort	

3rd person accusative clitics in Spanish/Italian project fixed node directly

IF	$?Ty(t), Tn(a)$	
THEN	IF	$[\downarrow_1^+]?Ty(x)$
	OR	
	IF	$Mood(-FINITE)$
	THEN	$\texttt{make}(\langle\downarrow_1\rangle); \texttt{go}(\langle\downarrow_1\rangle);$
		$\texttt{make}(\langle\downarrow_0\rangle); \texttt{go}(\langle\downarrow_0\rangle)$
		$\texttt{put}(Ty(e), Fo(U_{x'}), ?\exists \mathbf{x}.Fo(\mathbf{x}));$
		gofirst(?Ty(t))
ELSE	Abort	

The DS account of the Person Case Constraint (PCC)

- Given these lexical entries the PCC is explained via a treelogic constraint that bans the projection of more than one unfixed node

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After parsing *me* in *me-te* (Spanish):

$$Tn(a), \ldots ?Ty(t), \Diamond$$

$$\langle \uparrow_1^* \rangle Tn(a), ?\mathrm{Ty}(\mathrm{x})$$

$$\langle \uparrow_0 \rangle \langle \uparrow_1^* \rangle Tn(a),$$
$$Ty(e), Fo(U_{Speaker'})$$
$$?\exists \mathbf{x}.Fo(\mathbf{x}), ?\exists \mathbf{x}.Tn(\mathbf{x}).$$

The DS account

After parsing *te* in *me-te* (Spanish):

$$Tn(a), ...?Ty(t), \Diamond$$

$$\langle\uparrow_1^*\rangle Tn(a), ?\mathrm{Ty}(\mathrm{x})$$

$$\langle\uparrow_0\rangle\langle\uparrow_1^*\rangle Tn(a),$$
$$Ty(e), Fo(U_{Spkr'}), Fo(U_{Hr'})$$
$$?\exists\mathbf{x}.Fo(\mathbf{x}), ?\exists\mathbf{x}.Tn(\mathbf{x}).$$

- The two nodes collapse into one, carrying incompatible Formula values. The parse cannot be successful.

  Note: speakers find the debarred clusters wholly ill-formed

The DS account

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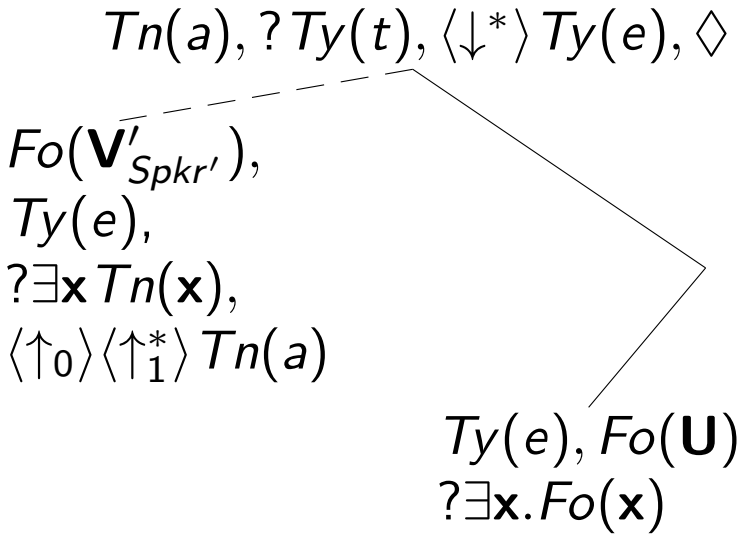
Parsing the clitic cluster me lo (Spanish)



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(72)

A: **Tu** milise xtes... B: Tu Giorgu (?)
A: him.CL-DAT talked-3SG yesterday... B: the-DAT George-DAT
' A: Yesterday he talked to him. B: to George (?)'

(73)

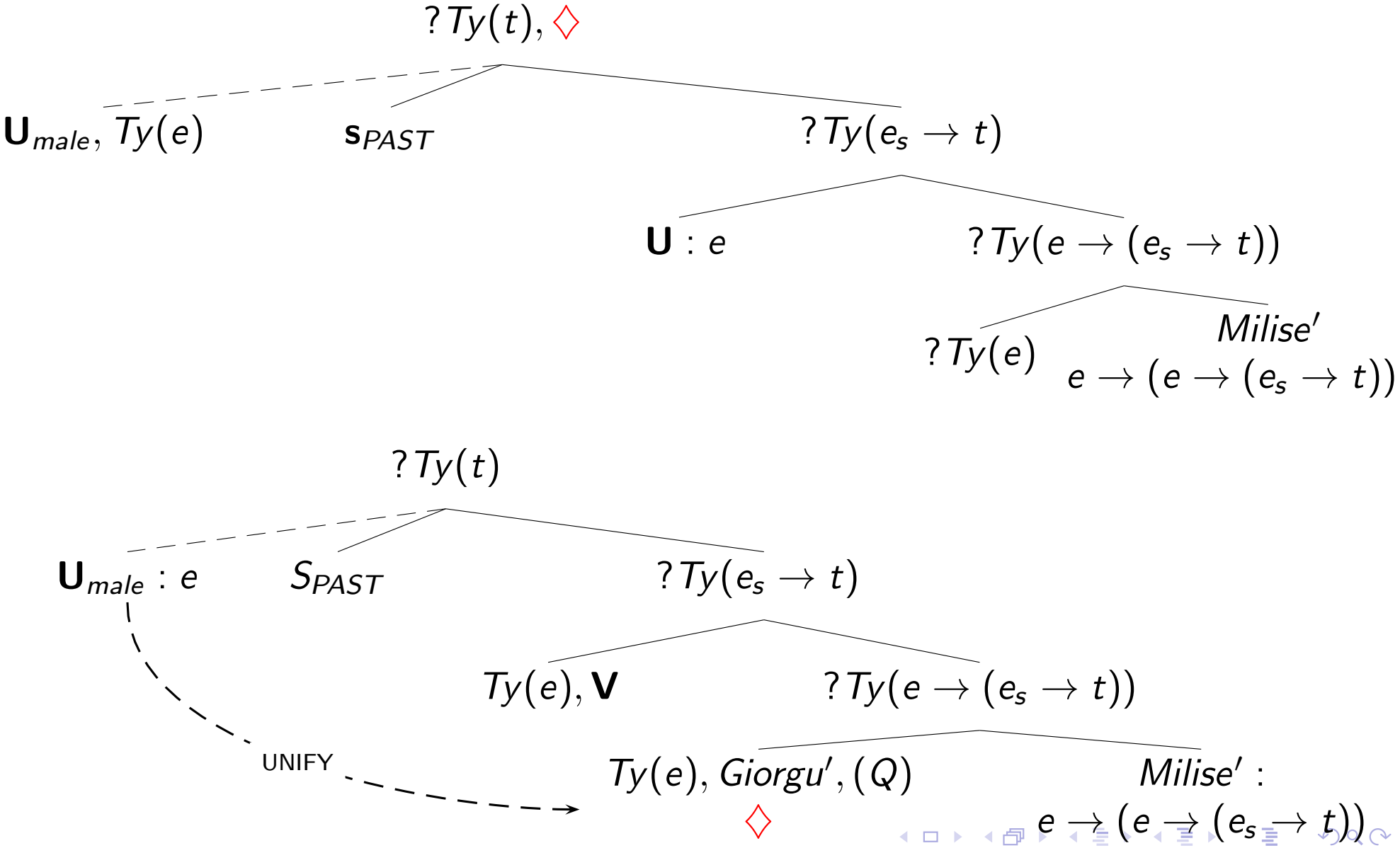
A: Tu Giorgu oposdipote B: **Tu** milise (?)
A: the-DAT George-DAT definitely B: him.CL-DAT talked-3SG
' A: To George definitely, B: He spoke to him(?)'

{A seeing John talking to Mary}

(74)

A: **Tis** ... B: milai tora (?), ne
A: her.CL-DAT ... B: talking-to-3SG now (?), yes
'A: To her ... B: he talks to her now, yes. Now he's talking to her? yes'

Parsing A: **Tu** milise xtes...B: Tu Giorgu (?)



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(76)

A:	Irthe	xtes	o	Giorgos	ke	tis...	B: to	edose?
A:	came	yesterday	the	George	and	her.CL-DAT	B: it.CL-ACC	gave

'A: Yesterday George came and to her ... B: he gave it her (?)'

- The question here is what is the nature of clitic clusters

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  - Are they comprised of individual lexical entries or a single routinized macro of actions is at play for the whole cluster?

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- There are cases of phonological reduction of some clusters in Greek involving the 2nd person dative clitic and third person accusative clitics
  a. su to → sto (to you it)
  b. su tin → stin (to you her)

Clitics in Split Utterances

- Can the split happen in these cases?
  - It seems not

(79)

??A: Irthe xtes o Giorgos ke s... B: to edose?
A: came yesterday the George and you.CL-DAT B: it gave
'A: Yesterday George came and to you ... B: he gave it her (?)'

- This would suggest that the split cannot happen inside a single lexical unit

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??A: Irthe xtes o Giorgos ke s... B: to edose?
A: came yesterday the George and you.CL-DAT B: it gave
'A: Yesterday George came and to you ... B: he gave it her (?)'

- This would suggest that the split cannot happen inside a single lexical unit
  - Only completed actions can be re-used (e.g. a full run of a computational or a lexical action)

Turn taking "inside" words

- But is this in general the case?
  - Can the split happen in between words?
    - It seems it can in some cases

(81)
A: Mas ipe pio itan to dia... B: kivevma?
A: us told what was the stake(half) B: stake
'A: He told us what was the B: stake'

(82) A: The DS people talk about meta...meta... B: variables?

- How should we approach these cases?

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  1. Interlocutors can actually take over in the middle of an action
  2. Interlocutors actually re-run the whole action (in a sense there is a phonology-syntax/semantics mismatch)
  3. Individual words might be comprised out of shorter individual actions.

Turn taking "inside" words

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(85)

??A:	Mas	ipe	pio	itan to	diak...	B: ivevma?
A:	us		told	what was	the stake(half)	B: stake

'A: He told us what was the B: stake'

- Of course this is a strange example, as there is a split in the middle of a syllable

(86)

?A:	Mas	ipe	pio	itan to	diaki...	B: vevma?
A:	us		told	what was	the stake(half)	B: stake

'A: He told us what was the B: stake'

- This is better but still degraded

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  - Let us think of a lexical entry for a word with actions A, that is split to parts B and C in the split:

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THEN	A
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- What happens if we take the second view is that the interlocutor contributing C is actually re-running the whole action and treats C as $C + B$, i.e. A
  - This requires a some kind of mismatch with phonology (and reminds us that we urgently need a phonological theory in DS)

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  - It is difficult to control such a case. Also it is very hard to have an exact correspondence between parts of the word and actions that will be clear
    - This seems like an infeasible task

Turn taking "inside" words

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  - Such chunks might be directly linked to their morphology
    - Turn taking can happen between these chunks
  - This assumption for words in DS is not new
    - Japanese suffixes were encoded as providing discrete action chunks reflecting constructive case properties (Cann et al. 2005)
- The entry for the Japanese accusative suffix -o (Cann et al. 2005:201)

IF	$?Ty(e)$	
THEN	IF	$[\uparrow^*](?Tn(a) \wedge ?Ty(t))$
	THEN	$\mathtt{put}(?\langle\uparrow_0\rangle(Ty(e \rightarrow t))$;
	ELSE	Abort
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  - In the condensed version *sto*, the cluster can be seen as involving one chunk of actions

Turn taking "inside" words

- Entry for *sto*

IF $?Ty(t), Tn(a)$
THEN IF $[\downarrow_1^+]?Ty(x)$
OR
IF $Mood(-FINITE)$
THEN $\mathtt{make}(\langle\downarrow_1\rangle); \mathtt{go}(\langle\downarrow_1\rangle)$
$\mathtt{make}(\langle\downarrow_1\rangle); \mathtt{go}(\langle\downarrow_1\rangle);$
$\mathtt{make}(\langle\downarrow_0\rangle); \mathtt{go}(\langle\downarrow_0\rangle);$
$\mathtt{put}(Ty(e), Fo(U_{Hr'}), ?\exists\mathbf{x}.Fo(\mathbf{x})); gofirst(?Ty(t));$
$\mathtt{make}(\langle\downarrow_1^*\rangle); \mathtt{go}(\langle\downarrow_1^*\rangle);$
$\mathtt{make}(\langle\downarrow_0\rangle); \mathtt{go}(\langle\downarrow_0\rangle);$
$\mathtt{put}(\langle\uparrow_0\rangle\langle\uparrow_1^*\rangle Tn(a));$
$\mathtt{put}(Ty(e), Fo(V_{neut'}), ?\exists\mathbf{x}.Fo(\mathbf{x}), \mathtt{go}(\langle\uparrow_0\rangle\langle\uparrow_1\rangle));$
$gofirst(?Ty(t))$
ELSE abort
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- For *su to* (to you it), the assumption is that this complex entry comes into two chunks
  - Entry for *su* (to you)

$$
\begin{array}{lll}
\text{IF} & ?Ty(t), Tn(a) & \\
\text{THEN} & \text{IF} & [\downarrow_1^+]?Ty(x) \\
 & \text{OR} & \\
 & \text{IF} & Mood(-FINITE) \\
 & \text{THEN} & \mathtt{make}(\langle\downarrow_1\rangle); \mathtt{go}(\langle\downarrow_1\rangle) \\
 & & \mathtt{make}(\langle\downarrow_1\rangle); \mathtt{go}(\langle\downarrow_1\rangle); \\
 & & \mathtt{make}(\langle\downarrow_0\rangle); \mathtt{go}(\langle\downarrow_0\rangle); \\
 & & \mathtt{put}(Ty(e), Fo(U_{Hr'}), ?\exists \mathbf{x}.Fo(\mathbf{x})); \\
 & & gofirst(?Ty(t));' \\
 & \text{ELSE} & \text{abort} \\
\text{ELSE} & \text{abort} &
\end{array}
$$

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  - Entry for *to* (to you)

$$
\begin{array}{lll}
\text{IF} & ?Ty(t), Tn(a) & \\
\text{THEN} & \text{IF} & [\downarrow_1^+]?Ty(x) \\
 & \text{OR} & \\
 & \text{IF} & Mood(-FINITE) \\
 & \text{THEN} & \mathtt{make}(\langle\downarrow_1^*\rangle); \mathrm{go}(\langle\downarrow_1^*\rangle); \\
 & & \mathtt{make}(\langle\downarrow_0\rangle); \mathrm{go}(\langle\downarrow_0\rangle); \\
 & & \mathtt{put}(\langle\uparrow_0\rangle\langle\uparrow_1^*\rangle Tn(a)); \\
 & & \mathtt{put}(Ty(e), Fo(V_{neut'}), ?\exists \mathbf{x}.Fo(\mathbf{x}), \mathrm{go}(\langle\uparrow_0\rangle\langle\uparrow_1\rangle)); \\
 & & gofirst(?Ty(t)) \\
 & \text{ELSE} & \text{abort} \\
\text{ELSE} & \text{abort} &
\end{array}
$$

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- With these assumptions in place, examples like the following are not a problem for DS

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(88)

A: Irthe xtes o Giorgos ke tis... B: to edose?
A: came yesterday the George and her.CL-DAT B: it gave
'A: Yesterday George came and to her ... B: he gave it her (?)'

- B can take over no matter what we take clitic clusters to involve (two separate entries or one entry with discrete action chinks)

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- In illicit cases (e.g. involving clitic cluster sto), the idea is that the cluster is comprised of one big chunk so the split is not possible

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(90)

??A: Irthe xtes o Giorgos ke s... B: to edose?
A: came yesterday the George and you.CL-DAT B: it gave
'A: Yesterday George came and to you ... B: he gave it her (?)'

The Person Case Constraint in Split Utterances

- The same person restrictions are operative in split utterances

(91)

A: Irthe xtes o Giorgos ke tu... B: *se edixe?
A: came yesterday the George and him B: *you showed (?)
‘ A: Yesterday George came and to him ... B: *he showed you to him (?)’

- When speaker B takes over, a locally unfixed node is present, projected via parsing the dative clitic. Parsing of the second clitic, will involve the projection of a second locally unfixed node.
  - The two nodes collapse into being the same node
    - The incompatible formula decorations on the same node will render the sentence ungrammatical

Conclusions

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Conclusions

- Clitics in split utterances present us with a challenge w.r.t. the status of clitic clusters but also clitics themselves
  - Turn taking can happen inside clitic clusters and sometimes even between words
  - The way to deal with this is to assume that either a some kind phonology-syntax/semantics interface is at play or: a) clitic clusters involve separate lexical entries or b) the complex entry comes into discrete action parts
- Person restrictions associated with clitic clusters are correctly predicted in the same sense as in the non-dialogue cases